



DILNI WIJESINGHE

BSc (Hons) Computer Science Undergraduate

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LinkedIn

ABOUT ME

Motivated Computer Science undergraduate at the Informatics Institute of Technology (IIT), affiliated with the University of Westminster, with a strong foundation in algorithm design, object-oriented programming, and applied machine learning. Experienced in building full-stack web applications and developing data-driven solutions using Java, Python, and SQL. Proven ability to translate complex requirements into technical solutions through Agile frameworks and robust source code design. Passionate about solving real-world challenges through software craftsmanship and seeking an Internship as a Software Engineer to contribute to high-performance development teams.

EDUCATION

BSc (Hons) Computer Science

Westminster University. (Sep 2024 – Present)

- Modules: Machine Learning, Database Systems, Algorithms, Web Application Development, Client-Server Architecture, OOP, Software Development.

GCE Advanced Level (Biological Science)

Mahanama National College Monaragala. (February 2024)

- Biology - C
- Chemistry - B
- Physics - S

TECHNICAL SKILLS

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|------------------------------|--|
| • Languages: | Python, Java, SQL |
| • AI & Machine Learning: | Supervised Learning, Deep Learning, Object Detection |
| • ML Frameworks & Libraries: | TensorFlow, OpenCV, YOLOv3, Darknet |
| • Databases & Data Formats: | MySQL |
| • Web Development: | HTML, CSS |
| • Design & Prototyping: | Figma, Canva |
| • Tools & Platforms: | GitHub, Postman, Kaggle, HackerRank |

ACHIEVEMENTS.

Collaborated as part of RobotDynomus Team to design and develop an autonomous maze-solving robot for the Micromouse Competition (IIT RAS), implementing the Left-Hand Rule algorithm in C++ for real-time navigation, contributing to a Top 10 team placement.

CAREER INTEREST

Artificial Intelligence, Machine Learning, Data Science, Software Engineering, Business Analysis

PROJECTS

University Projects

1. Project Coordinator & AI Developer | Harvesting Robot Arm (Harvestro Group Project)

Python, TensorFlow, OpenCV, YOLOv3-tiny, Raspberry Pi, Computer Vision

Developing an AI-powered agricultural robot that detects and harvests scotch bonnet chillies using computer vision and robotic automation. The system integrates deep learning-based object detection with a robotic arm to enable autonomous crop harvesting and sorting.

- Trained a YOLOv3-tiny object detection model for real-time chilli detection using annotated datasets.
- Implemented the image processing and detection pipeline using Python and OpenCV.
- Integrated the detection system with a Raspberry Pi-controlled robotic arm for automated harvesting.
- Collaborated within an Agile development environment, managing sprint tasks and tracking progress using ClickUp.
- Coordinated task allocation and monitored project milestones to ensure efficient team collaboration.

2. Climate Action Website - (EcoSpark - 2024)

HTML, CSS, JavaScript, Responsive Web Design

Developed a climate-action awareness website aligned with the United Nations Sustainable Development Goal (SDG 13 – Climate Action) to promote environmental responsibility and global collaboration.

- Collaborated in a team to develop an 8-page responsive climate action website.
- Built key pages using HTML, CSS, and JavaScript to present environmental data clearly.
- Worked with team members to integrate components and deliver the project successfully.

3. Data Analyst | Traffic Data Analysis Application (2024)

Python, Pandas, NumPy, Matplotlib, Data Visualization

Developed a traffic data analysis application to analyze transportation datasets and identify patterns in traffic flow.

- Collected and processed traffic datasets using Python and Pandas.
- Performed statistical analysis to identify traffic trends and patterns.
- Created data visualizations and charts using Matplotlib to present insights clearly.
- Generated reports summarizing findings to support data interpretation and decision-making.

4. Business & Domain Researcher | Cybersecurity Career & Industry Study (2024)

Cybersecurity Research, Threat Analysis, Industry Study

Conducted an exploratory study on cybersecurity careers and industry practices through an analyst interview and independent research.

- Investigated current cybersecurity threats, vulnerabilities, and defense strategies.
- Analyzed industry trends and career pathways in cybersecurity.
- Documented key insights related to threat mitigation and emerging security technologies.

5. Machine Learning Engineer | Loan Approval & Limit Prediction System (2026)

Python, Pandas, NumPy, Scikit-learn, Ensemble Learning, Regression, Decision Trees

Developed an automated financial lending system to classify loan eligibility and predict maximum credit limits for approved applicants using machine learning.

- Cleaned and preprocessed raw datasets by handling missing values via mean/mode imputation, encoding categorical variables, and removing duplicate entries.
- Developed, evaluated, and optimized multiple classification models, including Logistic Regression, Gaussian Naive Bayes, and K-Nearest Neighbors (KNN).
- Engineered a Voting Classifier ensemble model that combined classification algorithms to achieve a peak loan approval prediction accuracy of 99.9%.
- Built and optimized pruned Decision Tree Regressors to forecast maximum allowed loan amounts, evaluated using MAE, RMSE, and R^2 scores.

Self Projects

6.Data Analyst | Air Quality, Weather & Respiratory Health Analysis (2024)

Python, Pandas, NumPy, Matplotlib, Data Analysis, Machine Learning

Conducted a data-driven analysis to study the relationship between environmental factors and respiratory health outcomes using real-world datasets.

- Performed data cleaning, preprocessing, and exploratory data analysis using Python and Pandas.
- Created data visualizations and correlation analysis to identify relationships between pollution, weather conditions, and hospital visits.
- Developed a linear regression model to predict respiratory-related hospital admissions.
- Generated analytical reports and visual insights to support data-driven public health decision-making.

IEEE LEADERSHIP, VOLUNTEERING & MEMBERSHIPS

Leadership & Event Management (IEEE Student Branch – IIT)

- Vice Chair – Delegate Handling Team: IEEE BotWars Competition (Ongoing)
- Standing Committee Member: Volunteer Management Committee (Since 2025)
- Delegate Handling Executive: IEEE Prestige, IX UI/UX, CodeSprint, and Sherlock 3.0 Competitions (2024 – Present)

Community Service & International Relations (Leo Club)

- Partnership & International Relations Committee Member: Leo Club of University of Ceylon Alumni (2026 – 2027)

Professional Memberships (Since 2024)

- IEEE Global Member (Since 2024)
- IEEE Societies: Robotics & Automation (RAS) | Computer Society (CS) | Communications Society (CIS))

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